YiRage (YPK) — 系统架构 · RuntimeFusion 数据流

Engine owns schedule + KV; RF owns fusion identity; BackendRegistry searches and executes the same backend

主数据流　　虚线框 = 架构分区

回退 / 反馈　　盒内 = 模块 + 职责

项目介绍

YiRage 是面向大模型推理的多后端算子编译与运行时系统，核心能力是计算图超优化和 RuntimeFusion。

系统离线搜索并验证后端特定融合计划，再将 FusionCapsule 注入 vLLM 或 SGLang 的逐步解码流程。

INFERENCE ENGINE · 调度与 KV 所有者

- **vLLM** — continuous batch · PagedAttention
- **SGLang** — RadixAttention · extend meta
- **Decode Metadata** — block_table · seq_lens · radix hit · SM budget · batch

RuntimeFusion · RF.step(inputs, meta)

- **RF.step** — select 0..N capsules per decode step
- **FusionCapsule** — MLP gate_up + down · yirage.core
- **FusionCapsule** — segment pipeline · gate_up → down
- **Native Fallback** — skip → engine native ops

回退

SEARCH & PLAN · 离线搜索与预热

- **superoptimize** — KN / TB graph search
- **FusionPlan** — μGraph · profile + fingerprint
- **MuGraph cache** — shape bucket restore
- **Ray Coordinator** — 可选分布式搜索
- **Serving Tiers** — seed_verify · full_tb · full_tb_ray

PYTHON API 与 BACKEND MANAGER

- **KNGraph / TBGraph** — transpiler → kernels
- **BackendRegistry** — search + execute · same backend
- **yirage.core** — Cython runtime
- **yirage.serving** — RF · Capsule · cert
- **MACA Config** — warp = 64 · mxcc
- **CPU MLIR JIT** — 可选

BACKEND IMPLEMENTATIONS · 后端实现

- **CUDA** — nvcc · warp 32
- **MACA** — mxcc · warp 64
- **CPU** — OpenMP · MKL
- **MPS** — Apple GPU
- **Ascend** — CANN
- **ROCm** — AMD GPU
- **Triton** — DSL kernels
- **+ extra** — TPU · XPU · FPGA · MLIR

HARDWARE TARGETS · 硬件目标

- **NVIDIA GPU**
- **MetaX C500/C550**
- **x86 / ARM CPU**
- **Apple Silicon**
- **Huawei Ascend**
- **AWS Neuron** — Intel XPU

主链：decode metadata → RF.step → FusionCapsule → BackendRegistry → 同一后端与硬件。